Mini Project: Physics Lab using C

Visualizing Fundamental Laws of Optics and Mechanics

Course: B.Tech First Year (CS-AIDS)

Project Team

Division 10

1262254103

Omkar Hirave

Ray Optics

1262253047

Mandar Ghule

Wave Interference

1262251147

Aditya Modi

Projectile Motion

1262253678

Utkarsh Pagar

Wave Superposition

1262253092

Aditya Sale

Polarization

Ray Optics: Spherical Lenses

Presented by Omkar Hirave

Ray Approximation

Study of light propagation using the "Ray" model. Light travels in straight lines and bends at interfaces between different media.

Convex Lenses

Converging lenses that bring parallel rays to a focus. Positive focal length indicates light is bent toward the optical axis.

Concave Lenses

Diverging lenses that spread parallel rays apart. Negative focal length indicates light is bent away from the optical axis.

Key Formulas: Lens Maker's Formula: $\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$ and Magnification: $m = \frac{v}{u}$

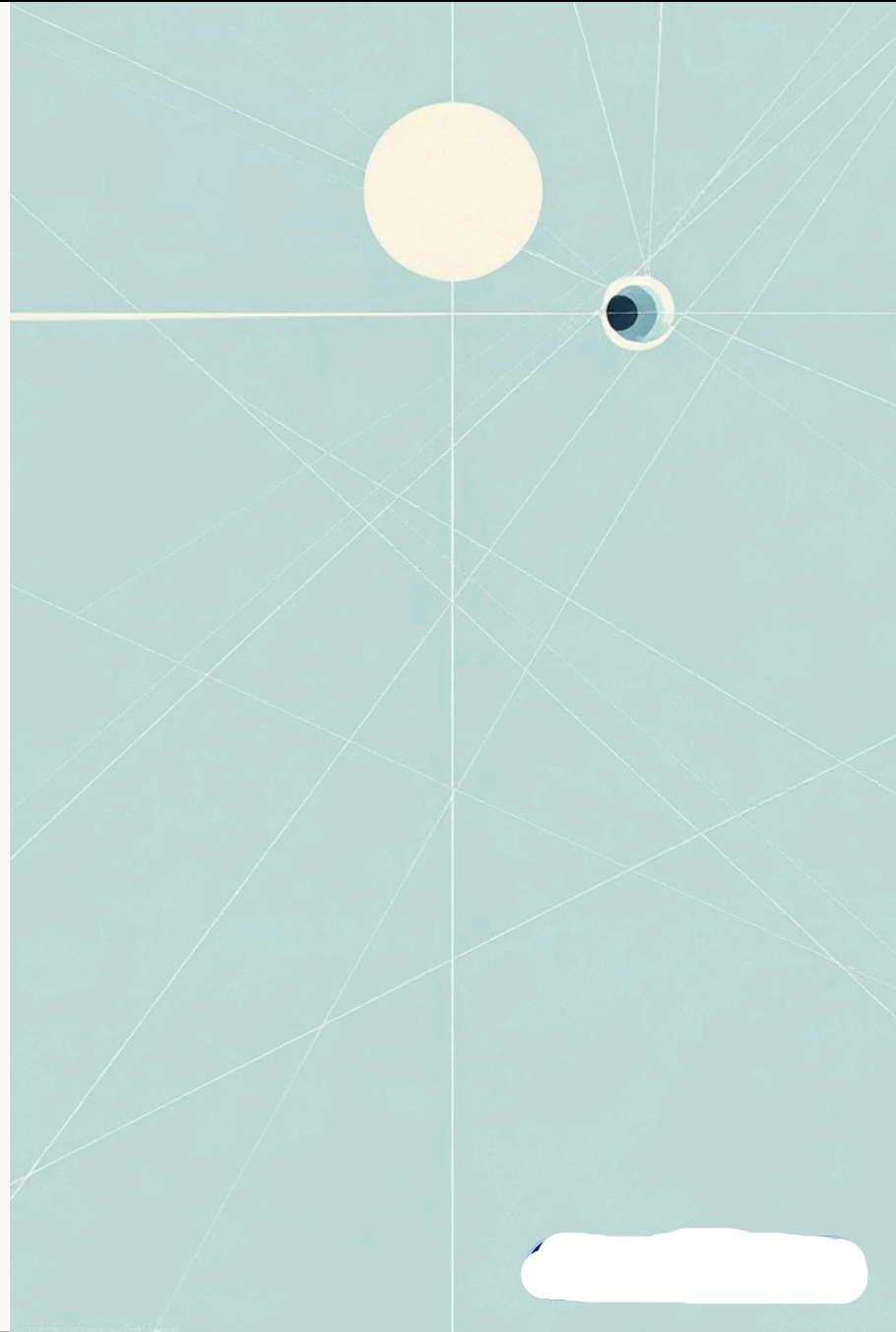


Image Formation Characteristics

Presented by Omkar Hirave

Analysis of image properties based on object position relative to focal length.

Real vs. Virtual

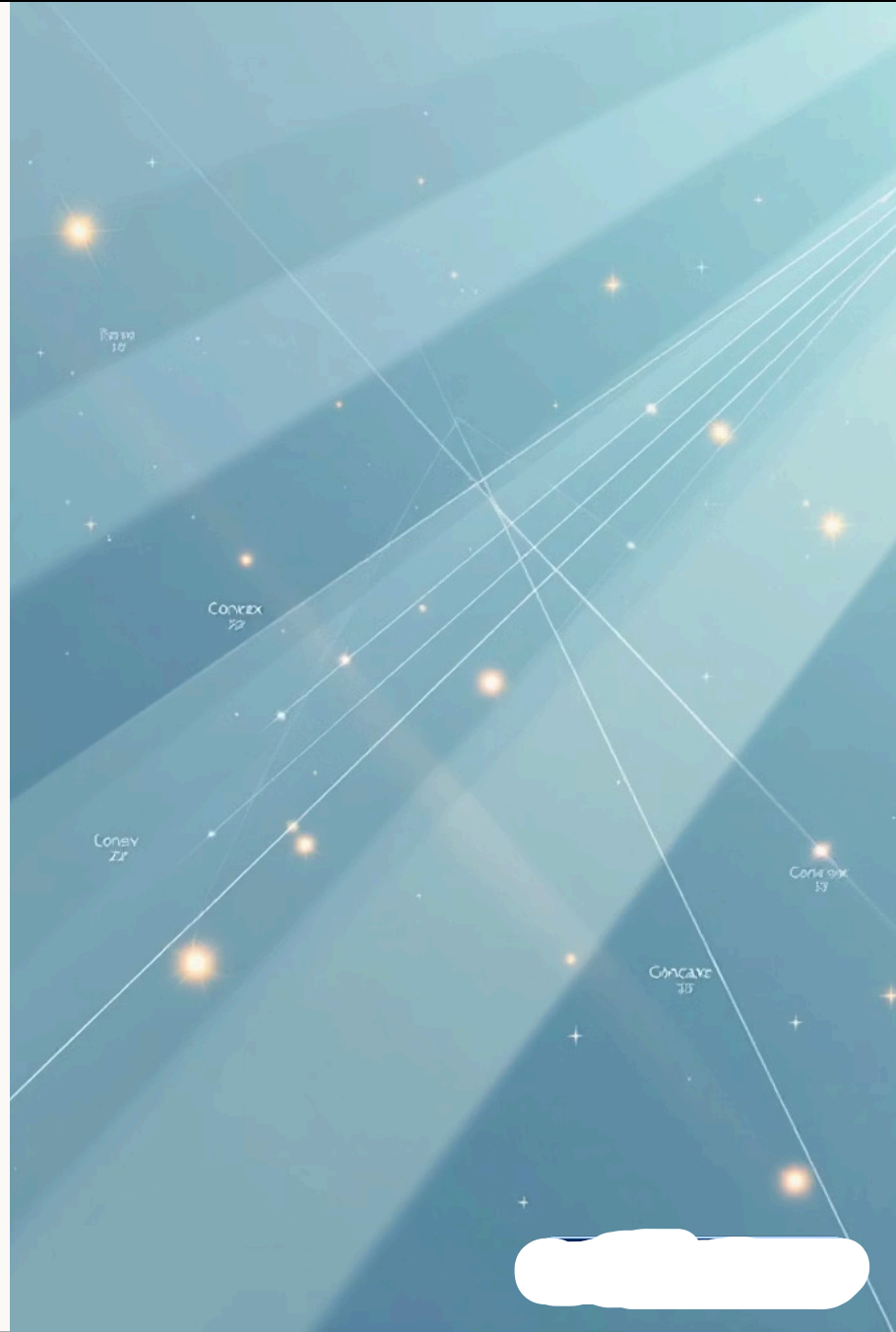
Real images form where rays converge (can be projected). Virtual images appear where rays diverge from (cannot be projected).

Inverted vs. Erect

Convex lenses produce inverted images when object is beyond focal point. Concave lenses always produce erect virtual images.

Lens Power

$P = \frac{1}{f}$ (measured in diopters when f is in meters). Converging lenses have positive power; diverging have negative power.



Wave Nature of Light: Interference

Presented by Mandar Ghule

Young's Double Slit Experiment demonstrates how coherent light sources create interference patterns through constructive and destructive superposition.

Constructive Interference

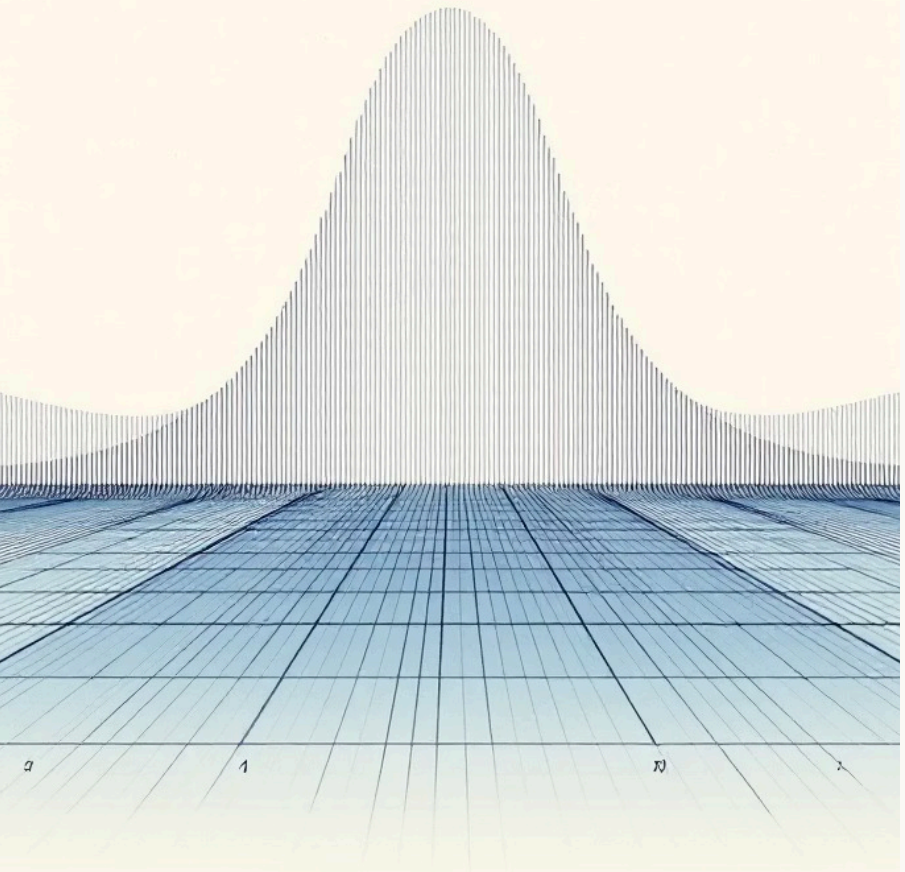
Path difference $\Delta x = n\lambda$ where n is an integer. Bright fringes appear where wave crests align.

Destructive Interference

Path difference $\Delta x = (n + \frac{1}{2})\lambda$. Dark fringes appear where crest meets trough.

$$\beta = \frac{\lambda L}{d}$$

Fringe width depends on wavelength (λ), slit separation (d), and screen distance (L)



Intensity Distribution

Presented by Mandar Ghule

Phase Difference

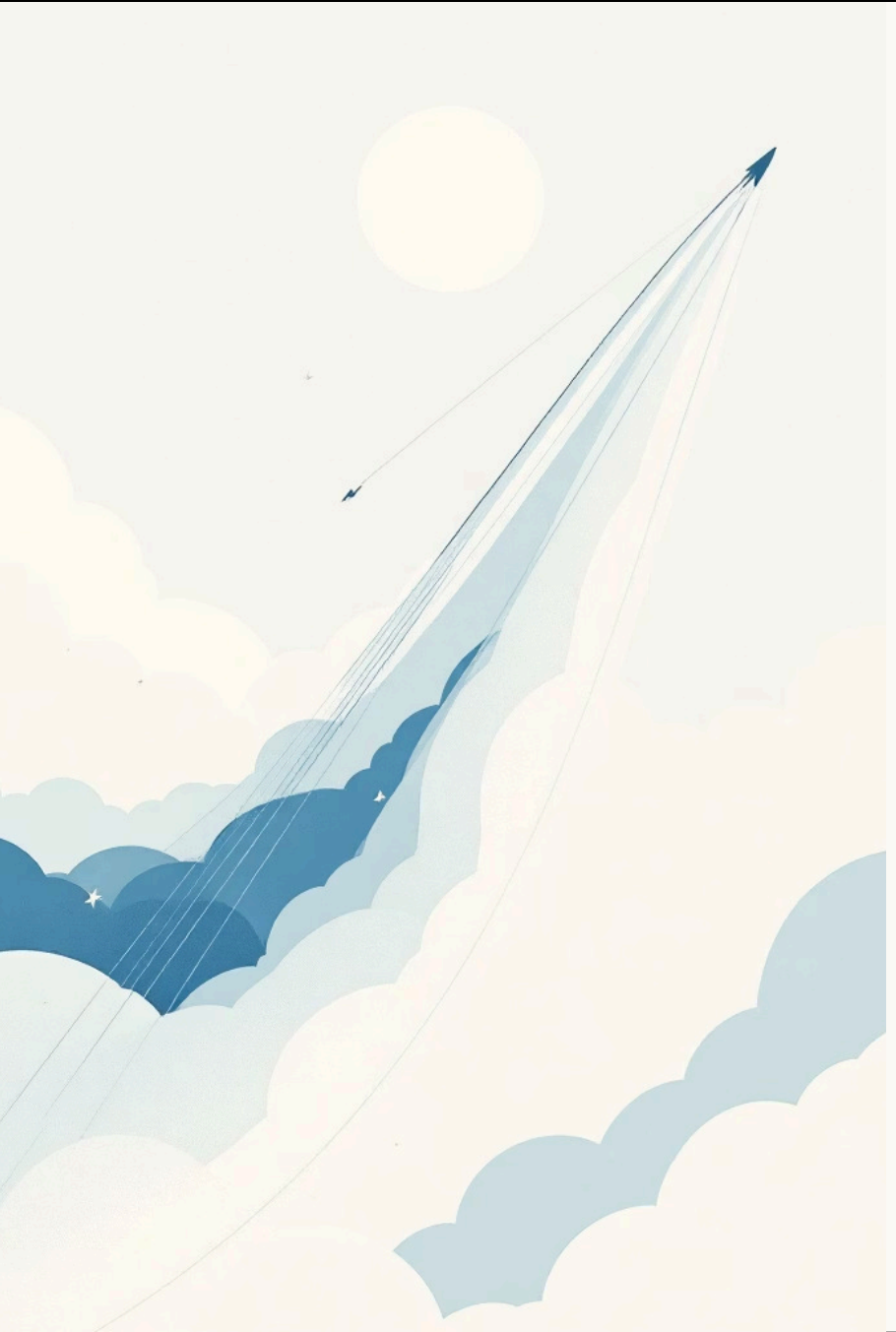
The relative shift between waves determines interference type. Zero phase difference produces maximum constructive interference.

Intensity Formula

Resultant intensity:
$I = I_1 + I_2 + 2\sqrt{I_1 I_2}\cos(\phi)$.
Maximum when φ = 2nπ, minimum when φ = (2n+1)π.

Coherence Required

Stable interference patterns only form with coherent sources having constant phase relationship and identical frequency.



Kinematics of Projectile Motion

Presented by Aditya Modi

1

Horizontal Motion

Constant velocity: $x = u\cos(\theta) \cdot t$. No acceleration in x-direction (neglecting air resistance).

2

Vertical Motion

Accelerated by gravity: $y = u\sin(\theta) \cdot t - \frac{1}{2}gt^2$. Symmetric parabolic trajectory.

3

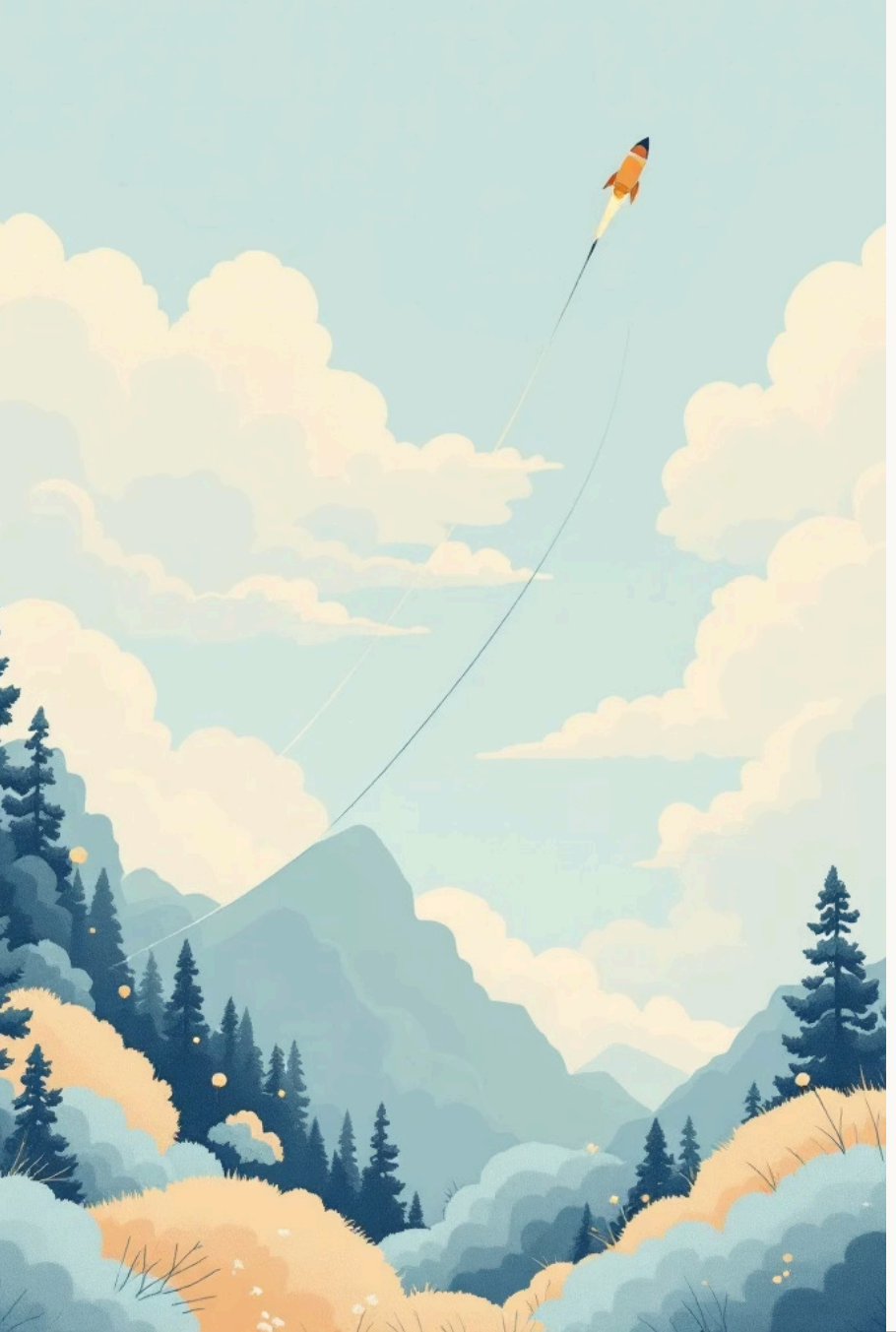
Range

Maximum horizontal distance: $R = \frac{u^2 \sin(2\theta)}{g}$. Maximum at θ = 45° launch angle.

4

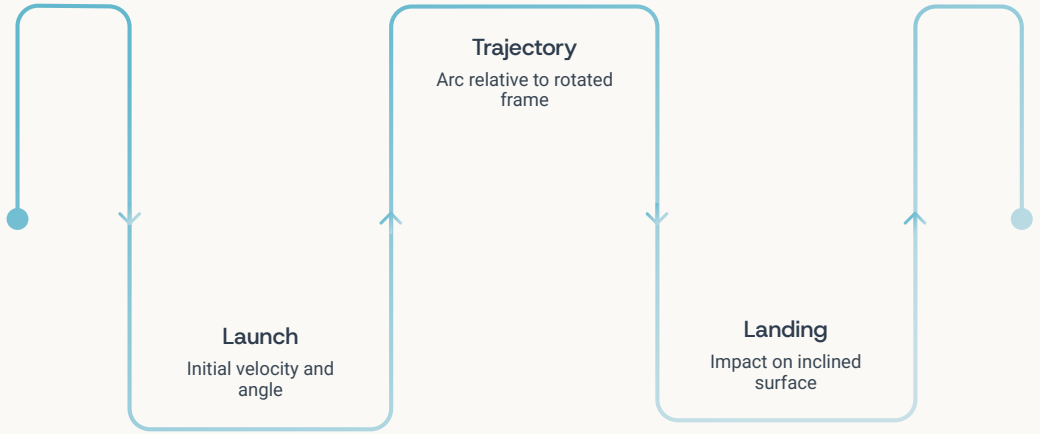
Maximum Height

Peak altitude: $H = \frac{u^2 \sin^2(\theta)}{2g}$. Achieved when vertical velocity becomes zero.



Motion on Inclined Planes

Presented by Aditya Modi



Modified Coordinate System

Reference frame rotates by angle α. Motion analysis adapts to slanted landing surface.

Time of Flight

Modified formula: $T = \frac{2u\sin(\theta-\alpha)}{g\cos(\alpha)}$. Landing time changes with incline angle.

Range on Incline

Distance along slope differs from horizontal range. Components parallel and perpendicular to surface analyzed separately.

Principle of Wave Superposition

Presented by Utkarsh Pagar

Core Concept

When multiple waves occupy the same space, their displacements add algebraically. The resultant wave determines observed interference patterns.

Constructive

Peaks align with peaks, producing amplified wave with maximum amplitude. Phase difference φ = 0, 2π, 4π...

Destructive

Peaks align with troughs, canceling wave completely. Phase difference φ = π, 3π, 5π...

$$y_{net} = A\sin(kx - \omega t + \phi)$$

Polarization of Electromagnetic Waves

Presented by Aditya Sale

Linear

Electric field oscillates in single plane. Simplest form where wave vibration direction remains constant.

Circular

E-field vector traces circle. Requires equal amplitudes with π/2 phase difference between perpendicular components.

Elliptical

Most general case where E-field traces ellipse. Occurs with unequal amplitudes or phase differences not exactly π/2.

Polarization restricts wave oscillations to specific orientations, crucial for applications in optics, communications, and imaging systems.

$$\Delta\phi = \frac{\pi}{2}, \quad A_x = A_y$$